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DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II

Advanced Calculus (2AC)

VECTOR CALCULUS III: The Green theorem

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0.1 Introduction

In this section we will deal with curves in the plane and also 2-dimensional vector fields. Before we present the main result in this section, we will need some few preliminary concepts.

Orientation of a curve: The concept of orientation of a curve is better understood when viewed from the physical context of the motion of an object along the curve. In that context, the orientation is viewed as the direction of the motion.

The concept of orientation of a curve is also easily understood through a notion of **order** that works only for simple non-closed curves. Precisely, if the simple non-closed curve is described by a vector function $\vec{r}: [a, b] \rightarrow \mathbb{R}^n$, then $\vec{r}(t')$ precedes $\vec{r}(t'')$ when $t' < t''$. In other words, an object moving along the curve passes first through $\vec{r}(t')$ then later through $\vec{r}(t'')$. So, a simple non-closed curve is oriented according to the increasing value of the parameter $t \in [a, b]$. However, this way of presenting the concept of orientation cannot be extended to simple closed curves since $\vec{r}(a) = \vec{r}(b)$. The following important theorem on closed simple curves in the plane is intuitively clear but its proof is known to be complex.

Theorem 0.1 (Jordan theorem) *Let $\mathcal{C}$ be a simple closed curve on the plane. Then the set $\mathbb{R}^2 \setminus \mathcal{C}$, the complement of the curve $\mathcal{C}$ is the union of two disjoint subregions of which one is unbounded and is called the **exterior of the curve $\mathcal{C}$** and the other which is bounded and is called the **interior of the curve $\mathcal{C}$** .*

Definition 0.2 (Orientation of a closed simple curve) A positive orientation of a simple closed curve $\mathcal{C}$ corresponds to a motion on the curve having the curve interior always to the left. Some textbooks like J. Stewart call it a **counterclockwise traversal** of the curve $\mathcal{C}$.

From what has been said in the section on unit tangent vector and principal unit normal vector in the chapter on vector functions and curves, the pair $(\vec{T}(t), \vec{N}(t))$ determines the positive orientation of the curve $\mathcal{C}$.

The following important theorem established a relation between a line integral $\int_{\mathcal{C}} \vec{F} \cdot d\vec{r}$ of a 2D vector field $\vec{F} = (P, Q)$ along a simple closed curve $\mathcal{C}$ and a double integral $\iint_D f(x, y) dx dy$, where D is the interior domain to the curve $\mathcal{C}$ and f is suitable function obtain from some partial derivatives of the components P and Q of the vector field $\vec{F}$.

0.2 The Green theorem

Theorem 0.3 (Green Theorem)

Let $\mathcal{C}$ be a piecewise regular simple closed curve with a counterclockwise orientation. Let D be the interior domain to the curve $\mathcal{C}$. Let U be subset of $\mathbb{R}^2$ which contains the domain D and the curve $\mathcal{C}$ and let $\vec{F} = (P, Q) : U \rightarrow \mathbb{R}^2$ be 2D vector field whose components are of class $\mathcal{C}^1$. Then

$$\oint_{\mathcal{C}} \vec{F} \cdot d\vec{r} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy \quad (1)$$

The identity (1) is referred to as the **Green identity** in some textbooks.

In general, the proof of Theorem 0.3 can involve some subtle difficulties. We will follow in these notes the general trend in many Calculus textbooks where the proof is presented in some special cases of the curve $\mathcal{C}$ and the domain D with the purpose of rather illustrating how the theorem works.

One can find the proof of the Green theorem for more complicated domains in advanced Analysis textbooks. **However, this is beyond the scope of this course. Below is a proof in a simple case but we leave it to the interested reader and we will not give any proof in the lectures. We will rather focus on how it is used.**

Proof of Theorem 0.3: The special case which is concerned with the descriptions of the curve $\mathcal{C}$ and its interior domain D is the following:

- (i) There are two functions $f_1, f_2 : [a, b] \rightarrow \mathbb{R}$ continuous such that $\mathcal{C}$ consists of two pieces, a lower piece

$$\mathcal{C}_1 := \{(x, y) \in \mathbb{R}^2 : y = f_1(x), \quad x \in [a, b]\}$$

and an upper piece

$$\mathcal{C}_2 := \{(x, y) \in \mathbb{R}^2 : y = f_2(x), \quad x \in [a, b]\}$$

and

$$D := \{(x, y) \in \mathbb{R}^2 : a \leq x \leq b, \quad f_1(x) \leq y \leq f_2(x)\}.$$

- (ii) There are other two functions $f_3, f_4 : [c, d] \rightarrow \mathbb{R}$ continuous such that $\mathcal{C}$ consists of two pieces, a left piece

$$\mathcal{C}_3 := \{(x, y) \in \mathbb{R}^2 : x = f_3(y), \quad y \in [c, d]\}$$

and a right piece

$$\mathcal{C}_4 := \{(x, y) \in \mathbb{R}^2 : x = f_4(y), \quad y \in [c, d]\}$$

and

$$D := \{(x, y) \in \mathbb{R}^2 : c \leq y \leq d, \quad f_3(y) \leq x \leq f_4(y)\}.$$

We want to show that

$$\oint_{\mathcal{C}} \vec{F} \cdot d\vec{r} = \oint_{\mathcal{C}} P(x, y)dx + Q(x, y)dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy \quad (2)$$

We will prove this in two steps:

Step 1: Using $\mathcal{C}$ and D described as in (i), let us show that

$$\boxed{\oint_{\mathcal{C}} P dx = - \iint_D \frac{\partial P}{\partial y} dx dy} . \quad (3)$$

Notice that

$$\begin{cases} \mathcal{C}_1 \text{ is described by } \vec{r}_1(x) := (x, f_1(x)) & x : a \rightarrow b, \\ \mathcal{C}_2 \text{ is described by } \vec{r}_2(x) := (x, f_2(x)) & x : b \rightarrow a. \end{cases}$$

We write

$$\begin{aligned} \oint_{\mathcal{C}} P dx &= \int_{\mathcal{C}_1} P(x, y) dx + \int_{\mathcal{C}_2} P(x, y) dx = \int_a^b P(x, f_1(x)) dx + \int_b^a P(x, f_2(x)) dx \\ &= \int_a^b [P(x, f_1(x)) - P(x, f_2(x))] dx \\ &= - \int_a^b \int_{f_1(x)}^{f_2(x)} \frac{\partial P}{\partial y}(x, y) dy dx = - \iint_D \frac{\partial P}{\partial y}(x, y) dx dy. \end{aligned}$$

Step 2: Now using $\mathcal{C}$ and D described as in (ii), let us show that

$$\boxed{\oint_{\mathcal{C}} Q dy = \iint_D \frac{\partial Q}{\partial x} dx dy} . \quad (4)$$

Notice that

$$\begin{cases} \mathcal{C}_3 \text{ is described by } \vec{r}_3(y) := (f_3(y), y) & y : d \rightarrow c, \\ \mathcal{C}_4 \text{ is described by } \vec{r}_4(y) := (f_4(y), y) & y : c \rightarrow d. \end{cases}$$

As before, we write

$$\begin{aligned} \oint_{\mathcal{C}} Q dy &= \int_{\mathcal{C}_3} Q(x, y) dy + \int_{\mathcal{C}_4} Q(x, y) dy = \int_d^c Q(f_3(y), y) dy + \int_c^d Q(f_4(y), y) dy \\ &= \int_c^d [-Q(f_3(y), y) + Q(f_4(y), y)] dy \\ &= \int_c^d \int_{f_3(y)}^{f_4(y)} \frac{\partial Q}{\partial x}(x, y) dx dy = \iint_D \frac{\partial Q}{\partial x}(x, y) dx dy. \end{aligned}$$

Therefore, we obtain (2) from (3) and (4) and this completes the proof of the theorem. ■

Remark 0.4 It follows from the Green theorem that one can use the line integral to evaluate a double and viceversa one can use the double integral to evaluate a line integral depending on the convenience. For instance, through the Green theorem, one can evaluate the area of a domain in $\mathbb{R}^2$. In fact, from the chapter on multiple integrals, we found that the area of a domain $D \subset \mathbb{R}^2$ is

$$A(D) = \iint_D 1 dx dy.$$

Now assume that the domain D and its boundary $\mathcal{C}$ satisfy the assumption the Green theorem (this could happen if D is the region enclosed by an ellipse), then we seek the functions P, Q such that

$$A(D) = \iint_D 1 dx dy = \oint_{\mathcal{C}} [P(x, y) dx + Q(x, y) dy]$$

provided the curve $\mathcal{C}$ has a counterclockwise orientation. In other words, we seek functions P and Q such that

$$\boxed{\frac{\partial Q}{\partial x}(x, y) - \frac{\partial P}{\partial y}(x, y) = 1 \quad \forall (x, y) \in D}.$$

We can choose the simple case where

$$\boxed{P(x, y) = -\frac{1}{2}y \quad \text{and} \quad Q(x, y) = \frac{1}{2}x}.$$

In that case we get that

$$\boxed{A(D) = \oint_{\mathcal{C}} -\frac{1}{2}y dx + \frac{1}{2}x dy} \quad (5)$$

One could also choose

$$\boxed{P(x, y) = 0 \quad \text{and} \quad Q(x, y) = x}.$$

For example, find the area $A(D)$ of the domain D enclosed by the ellipse E with cartesian equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad a, b > 0.$$

Using (5) we find that

$$A(D) = \oint_E -\frac{1}{2}y dx + \frac{1}{2}x dy.$$

Now, we can describe the ellipse E with the vector function

$$\vec{r}(t) := (a \cos t, b \sin t), \quad t \in [0, 2\pi] \quad \text{that is} \quad \begin{cases} x = a \cos t, \\ y = b \sin t, \end{cases} \quad t \in [0, 2\pi]$$

and get

$$\begin{aligned} A(D) &= \int_0^{2\pi} \left[-\frac{1}{2}y \frac{dx}{dt} + \frac{1}{2}x \frac{dy}{dt} \right] dt = \int_0^{2\pi} \left[-\frac{1}{2}(b \sin t)(-a \sin t) + \frac{1}{2}(a \cos t)(b \cos t) \right] dt \\ &= \frac{1}{2}ab \int_0^{2\pi} [\sin^2 t + \cos^2 t] dt = \frac{1}{2}(ab)2\pi = \pi ab. \end{aligned}$$

Now, let us check that this is true by evaluating directly the double integral using the techniques developped in the previous chapter on multiple integrals. In that case we make use of the change of variables

$$x = ar \cos \theta \quad \text{and} \quad y = br \sin \theta, \quad r \in [0, 1], \quad \theta \in [0, 2\pi]$$

with the Jacobian

$$J = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} a \cos \theta & -ar \sin \theta \\ b \sin \theta & br \cos \theta \end{vmatrix} = abr.$$

Hence,

$$A(D) = \iint_D 1 dx dy = \int_0^{2\pi} \int_0^1 a b r dr d\theta = \pi a b \int_0^1 2r dr = \pi a b [r^2]_0^1 = \pi a b .$$

Example 0.5 Let D be the domain enclosed by the line $y = x$ and the parabola $y = x^2$. Let $\vec{F}(x, y) = (x^2y, 2xy)$ for every $(x, y) \in \mathbb{R}^2$. Check that the identity if the Green theorem holds in this case.

Solution: Sketch the domain D !. Setting

$$P(x, y) := x^2y \quad \text{and} \quad Q(x, y) := 2xy$$

we get that

$$\begin{aligned} \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy &= \iint_D (2y - x^2) dx dy = \int_0^1 \int_{x^2}^x (2y - x^2) dy dx = \int_0^1 \left[y^2 - x^2y \right]_{x^2}^x dx \\ &= \int_0^1 (x^2 - x^3) dx = \left[\frac{1}{3}x^3 - \frac{1}{4}x^4 \right]_0^1 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}. \end{aligned}$$

Now, let us evaluate the line integral $\oint_C P(x, y)dx + Q(x, y)dy$ bearing in mind that the curve C with a counterclockwise orientation, is made of

$$C_1 : (x, x^2) \text{ with } x : 0 \rightarrow 1 \quad \text{and} \quad C_2 : (x, x) \text{ with } x : 1 \rightarrow 0.$$

Thus,

$$\begin{aligned} \oint_C P(x, y)dx + Q(x, y)dy &= \underbrace{\int_{C_1} P(x, y)dx + Q(x, y)dy}_{\text{Here } y = x^2} + \underbrace{\int_{C_2} P(x, y)dx + Q(x, y)dy}_{\text{Here } y = x} \\ &= \int_0^1 P(x, x^2)dx + Q(x, x^2)2xdx + \int_1^0 P(x, x)dx + Q(x, x)dx \\ &= \int_0^1 (x^4 + 4x^4)dx - \int_0^1 (x^3 + 2x^2)dx = \int_0^1 (5x^4 - x^3 - 2x^2)dx \\ &= \left[x^5 - \frac{1}{4}x^4 - \frac{2}{3}x^3 \right]_0^1 = 1 - \frac{1}{4} - \frac{2}{3} = \frac{1}{12}. \end{aligned}$$

Hence,

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \frac{1}{12} = \oint_C P(x, y)dx + Q(x, y)dy .$$

Exercises 0.6

1. Verify the Green theorem for each of the following:

- (a) $P(x, y) = \frac{y^2}{2}$, $Q(x, y) = \frac{x^2}{4}$ and D is the domain enclosed by the parabola $x = 9 - y^2$ and the y -axis with its boundary C having a counterclockwise orientation. (Hint: First describe and sketch the domain D indicating the counterclockwise orientation of its boundary C and then show that the identity in the Green theorem holds.)

Answer: $\boxed{\frac{324}{5}}.$

- (b) $P(x, y) = e^x \sin y$, $Q(x, y) = xy$ and D is the domain enclosed by the triangle with vertices at the points $(0, 0)$, $(2, 0)$ and $(1, 1)$. **Answer:** $\boxed{\frac{1}{3} - \int_0^1 (e^x + e^{2-x}) \sin x \, dx}.$

2. Using the Green Theorem, evaluate the line

$$\oint_C (e^{x^2} - y)dx + (e^y + x^4/4)dy$$

where C is the boundary curve of the domain

$$D := \{(x, y) \in \mathbb{R}^2: 9x^2 + y^2 \leq 9, x \geq 0, y \geq 0\},$$

with C having a counterclockwise orientation. **Answer:**

$$\boxed{\frac{2}{5} + \frac{3\pi}{4}}.$$

3. Given the vector field $\vec{F}(x, y) = (P(x, y), Q(x, y))$ with $P(x, y) = x \arctan y$ and $Q(x, y) = x$. Let $D := \{(x, y) \in \mathbb{R}^2: 0 \leq y \leq 1 - x^2\}$. Evaluate double integral $\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$ in two ways:

- (i) directly; (ii) using the Green theorem.

Answer: $\boxed{\frac{4}{3}}.$

4. Let $f: \mathbb{R} \rightarrow \mathbb{R}^+$ be a continuously differentiable function such that

$$[f(t)]^2 + [f'(t)]^2 > 0 \quad t \in \mathbb{R}.$$

Let $\mathcal{C}$ be a closed simple curve described in polar coordinates (r, θ) by the equation

$$r = f(\theta) \quad \text{with } \theta_1 \leq \theta \leq \theta_2.$$

- (a) Write the parametric equations for the curve $\mathcal{C}$.
 (b) Show that $\mathcal{C}$ is a regular curve.
 (c) Let D be the domain enclosed by the curve $\mathcal{C}$. Use The Green theorem to show that area of the domain D is equal to $\frac{1}{2} \int_{\theta_1}^{\theta_2} [f(\theta)]^2 d\theta$.

Hint: $A(D) = \iint_D 1 dx dy = \underbrace{\int_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \int_C P dx + Q dy}_{\text{Green Theorem: counterclockwise orientation for } C}$. So, choose P and Q

such that $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$. For instance, you can take $Q(x, y) = x$ and $P(x, y) = 0$ or take $Q(x, y) = 0$ and $P(x, y) = -y$ or take $Q(x, y) = \frac{x}{2}$ and $P(x, y) = -\frac{y}{2}$.

(d) Evaluate the area of D in the case $f(\theta) = 1 + \cos \theta$ with $\theta_1 = 0$ and $\theta_2 = 2\pi$.

Answer: $\boxed{\frac{3\pi}{2}}$.

**The next set of notes will be
on the Stokes theorem!**